METHODS

Study Design

- Cross-sectional analysis of HPP 10K cohort
- Sample size: N = 7,722 subjects with both sleep study and CGM data
- Age range: 40-70 years (per HPP dataset guidelines)

Exposure Variable

- Apnea-Hypopnea Index (AHI): events per hour of sleep
- Categorized as: Normal (<5), Mild (5-15), Moderate (15-30), Severe (>=30)

Outcome Variables (from Continuous Glucose Monitoring)

- MAGE: Mean Amplitude of Glycemic Excursions (primary outcome)
- Glucose CV: Coefficient of Variation
- Glucose SD: Standard Deviation
- Time in Range: % time glucose 70-180 mg/dL

Statistical Analysis

- Unadjusted: ANOVA across OSA severity groups
- Adjusted: Multiple linear regression controlling for:
  - Age (continuous)
  - Gender (binary)
  - BMI (continuous)
- Partial correlations to assess independent associations
- Stratified analysis by diabetes status (Normoglycemic vs Prediabetes)

Interpretation Guide

- Partial correlation (r): Strength of association after removing confounders
- Beta coefficient: Expected change in outcome per 1-unit increase in AHI
- Effect size reported as MAGE change for AHI increase of 25 (mild to severe)

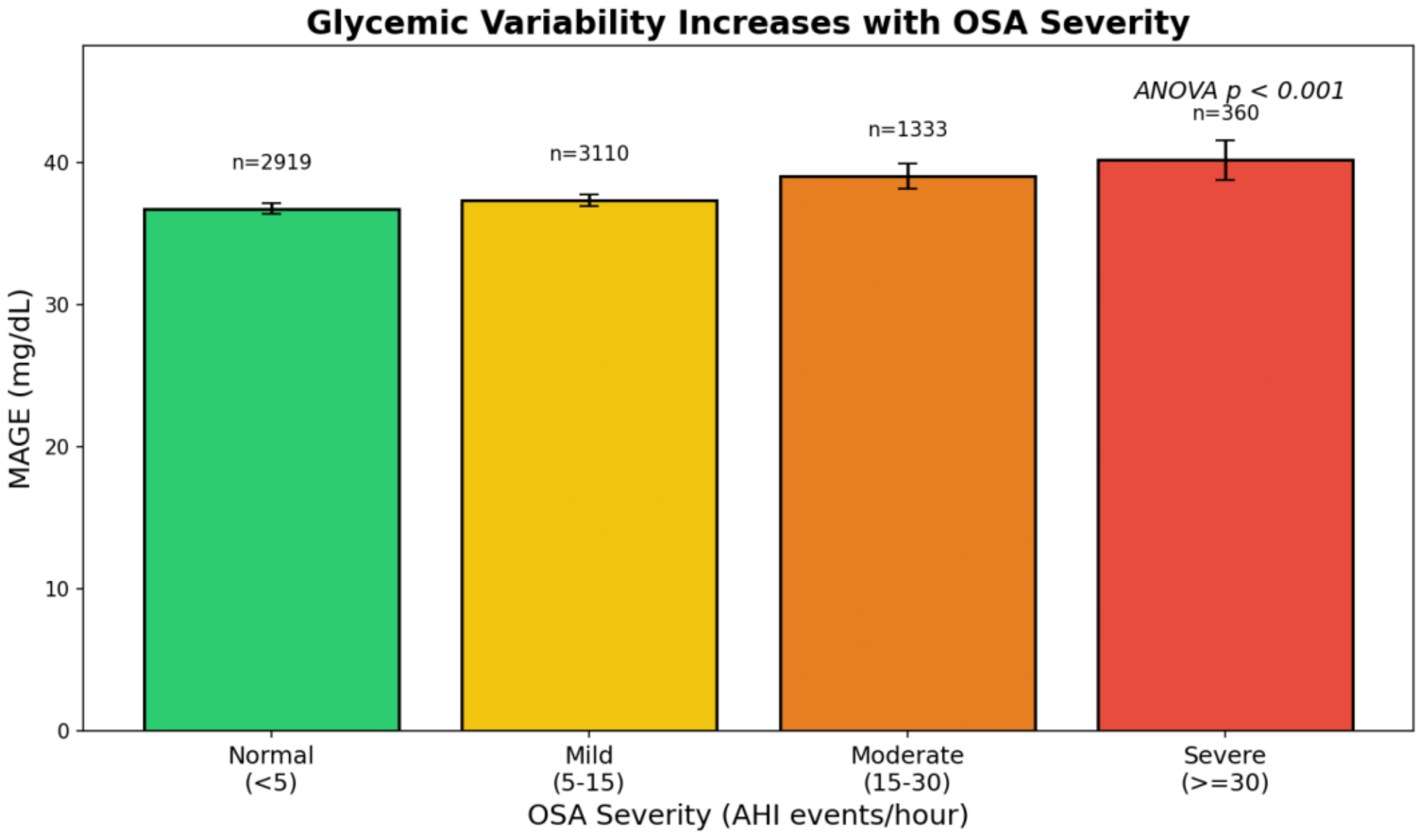
Inclusion Criteria

- Age 40-70 years
- Complete data on AHI and at least one glycemic metric
- No missing confounders (age, gender, BMI)

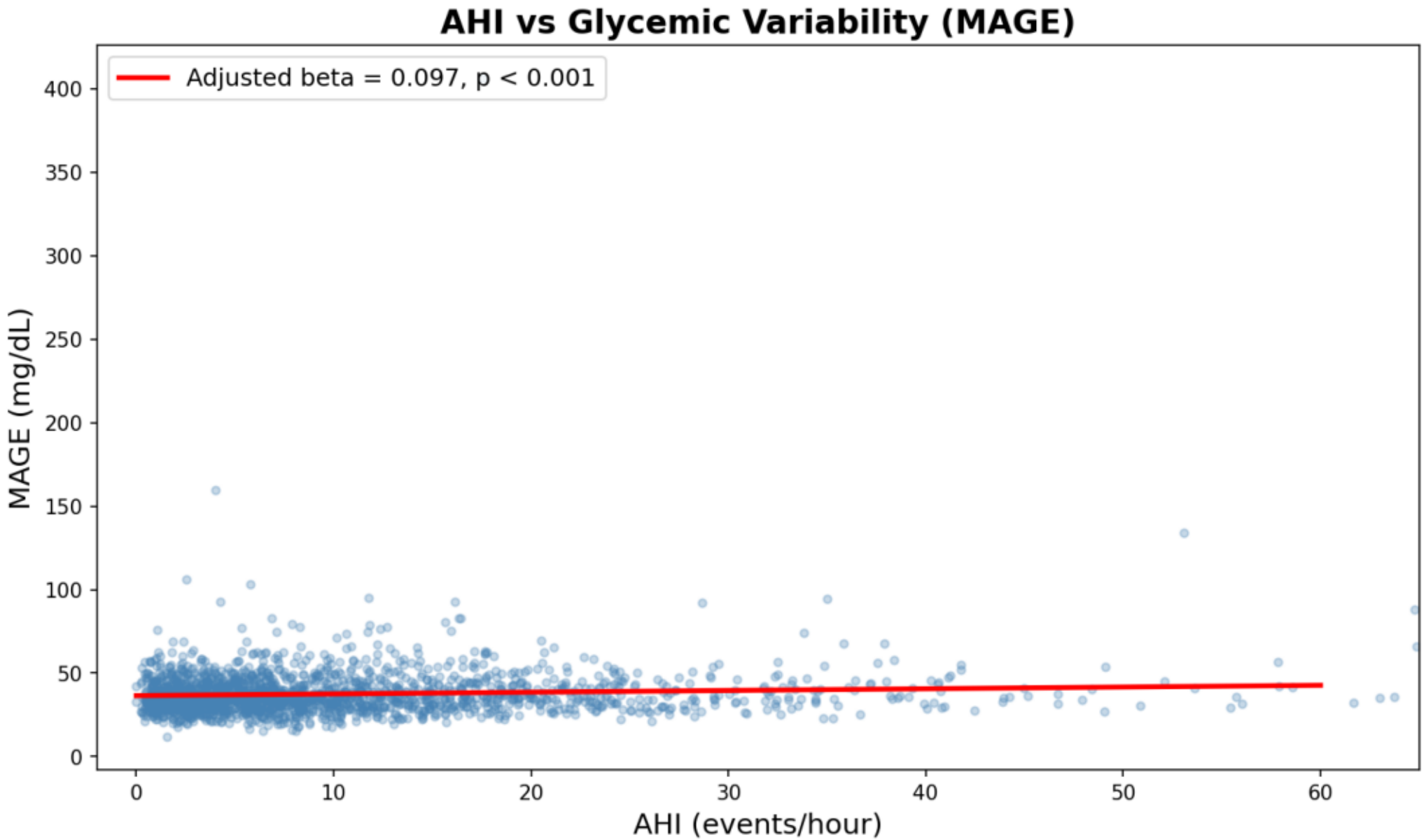
Software

- Python 3.x with statsmodels, scipy, pandas
- Analysis code: research/osa_glycemic_variability/analysis.py

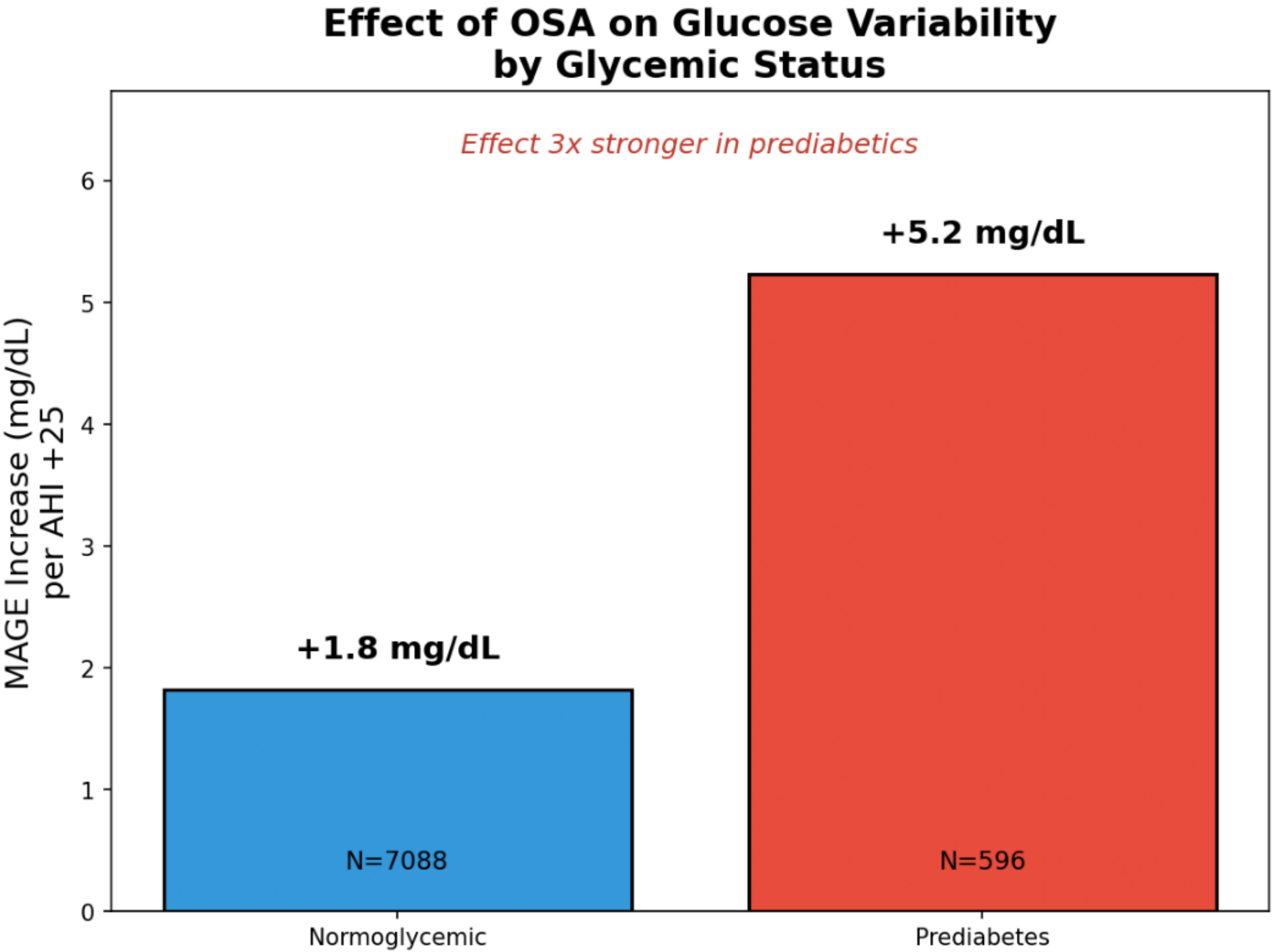
Main Finding: MAGE Increases with OSA Severity



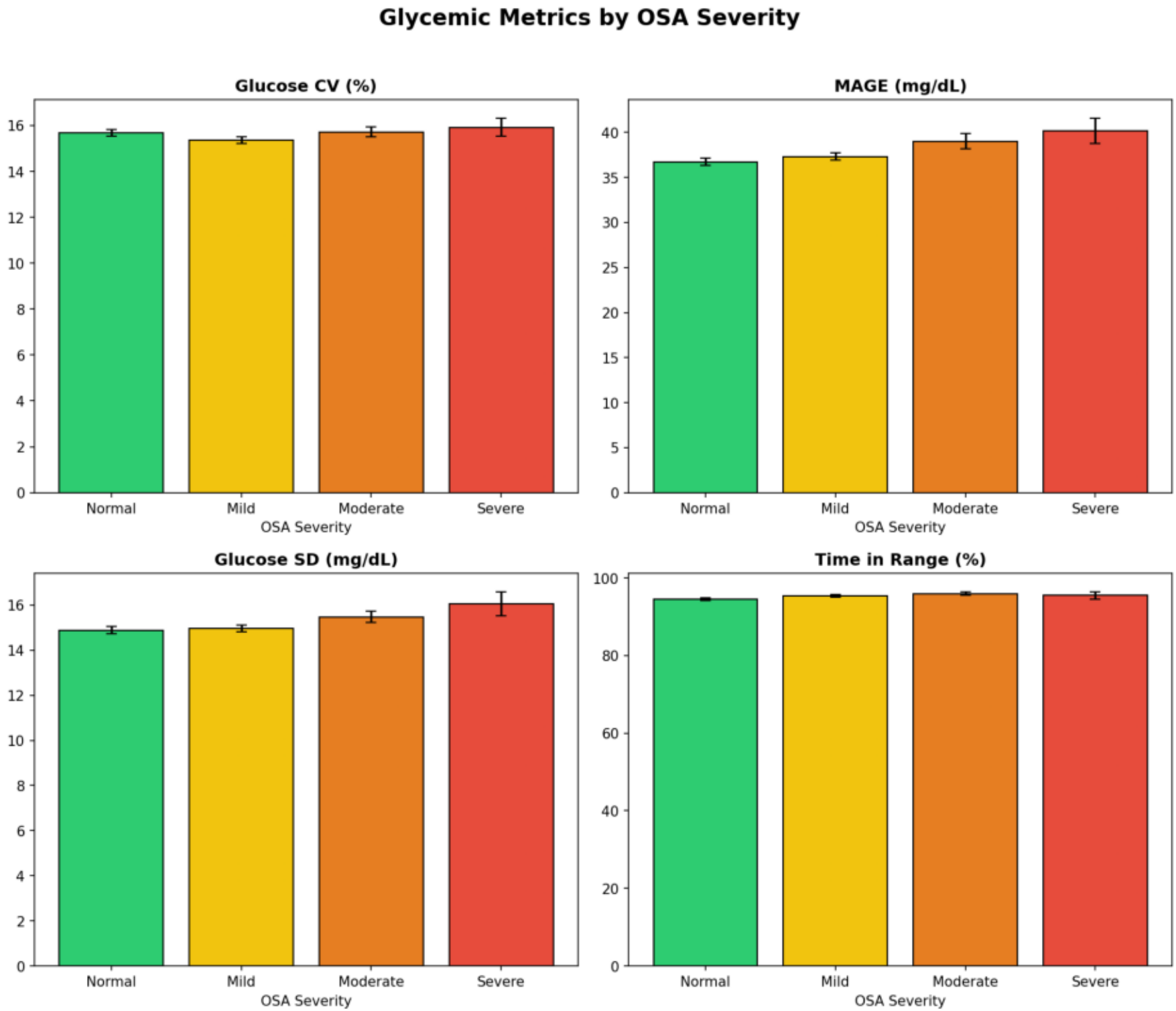
Continuous Association: AHI vs MAGE



Effect Modification: Stronger Effect in Prediabetics



All Glycemic Metrics by OSA Severity



KEY FINDINGS

Main Result

OSA severity (AHI) is significantly associated with glucose variability after controlling for age, gender, and BMI.

Partial Correlations (adjusted):

- AHI ~ MAGE: $r = 0.068$, $p < 0.001$ ***
- AHI ~ Glucose CV: $r = 0.043$, $p < 0.001$ ***
- AHI ~ Glucose SD: $r = 0.058$, $p < 0.001$ ***
- AHI ~ Time in Range: $r = -0.001$, $p = 0.94$ (NS)

Effect Size

Going from mild OSA (AHI=5) to severe OSA (AHI=30):

Overall:	+2.4 mg/dL MAGE (~6.5% increase)
Normoglycemic:	+1.8 mg/dL MAGE
Prediabetes:	+5.2 mg/dL MAGE (~14% increase)

>> Effect is 3x STRONGER in prediabetics <<

Clinical Implications

- OSA screening may be particularly important in prediabetes
- Sleep interventions could potentially improve glycemic stability
- Effect sizes are modest but significant at population level

Limitations

- Cross-sectional design: cannot establish causality
- Possible residual confounding (diet, physical activity, medications)
- BMI paradox observed: higher BMI associated with LOWER variability (unexplained, warrants further investigation)

Strengths

- Large sample size (N = 7,722)
- Objective sleep study measures (not self-reported)
- Continuous glucose monitoring (superior to HbA1c alone)
- Robust to multiple sensitivity analyses